BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 268 · Five process scheduling algorithms

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a Ready process?

Q2. What is a scheduling quantum?

Q3. What is a context switch?

Q4. Sorted [1,3,5,7,9,11,13,15,17], target 15. Trace midpoint indexes.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

FCFS — First Come First Served.

SJF — Shortest Job First.

SRT — Shortest Remaining Time.

RR — Round Robin.

MLFQ — Multilevel feedback queue.

New learning

Understand — FCFS serves arrival order; SJF selects the shortest available complete job.

Trace — SRT can preempt when a newly ready job has shorter remaining time; RR cycles by quantum.

Evaluate — Multilevel feedback queues adapt priority by task behaviour. Compare responsiveness, waits, starvation and context-switch overhead.

Worked example

A=4,B=2 arrive together: trace non-preemptive SJF.

1. Predict first — Shorter job B runs first

2. Model and check — A waits two time units.

We Do · Practise with guidance

A=3,B=2,C=1 all at time zero: trace FCFS and waits. Break simultaneous-arrival ties in the listed A, B, C order.

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A 0–3; B 3–5; C 5–6.
Wait A=0, B=3, C=5.
```

Predict first

Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.

Model and check

Explain why each value follows from the earlier step.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name FCFS,SJF,SRT,RR and multilevel feedback by defining feature. [5]

Question 2

A=5,B=3,C=1 all arrive at t=0. For FCFS, break ties in A, B, C order; for non-preemptive SJF, select the shortest available job. Compare the two orders and mean waiting times. [6]

Question 3

Using Round Robin with time quantum 2, processes A=4, B=3 and C=1 all arrive at t=0; take the initial queue order as A, B, C. Give all execution slices and one drawback of a short quantum. [6]

Exit Ticket · Answer before checking

Round Robin always lets one process finish before any other runs.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Eligible for CPU time.

Q2. Maximum CPU time per RR turn.

Q3. Saving and restoring execution context when changing process.

Q4. mid4→9, mid6→13, mid7→15 found; indexes 4,6,7.

You Do pages · Independent questions

Q1. Arrival order [1]; shortest available non-preemptive job [1]; preemptive shortest remaining [1]; fixed time quanta [1]; adaptive multiple queues [1].

Q2. FCFS A,B,C [1], waits 0,5,8 [1], mean13/3 [1]; SJF C,B,A [1], waits 0,1,4 [1], mean5/3 [1].

Q3. A0–2 [1], B2–4 [1], C4–5 [1], A5–7 [1], B7–8 [1]; small quantum adds context-switch overhead [1].

Exit Ticket · Model answer

False. It normally preempts each process after a time quantum.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: